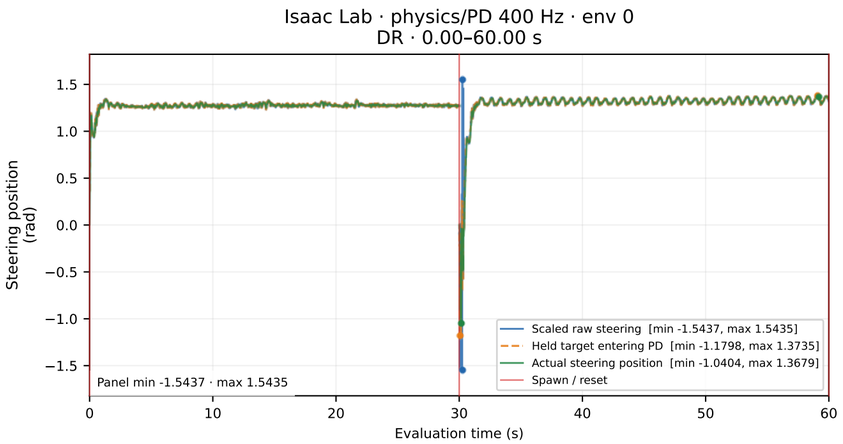
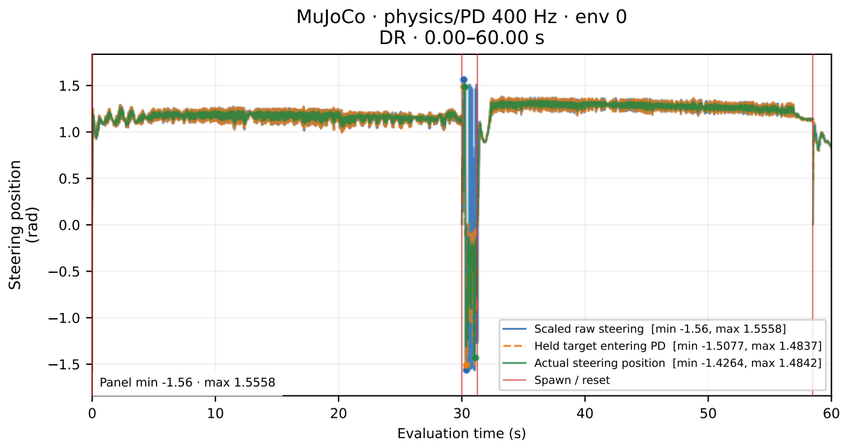
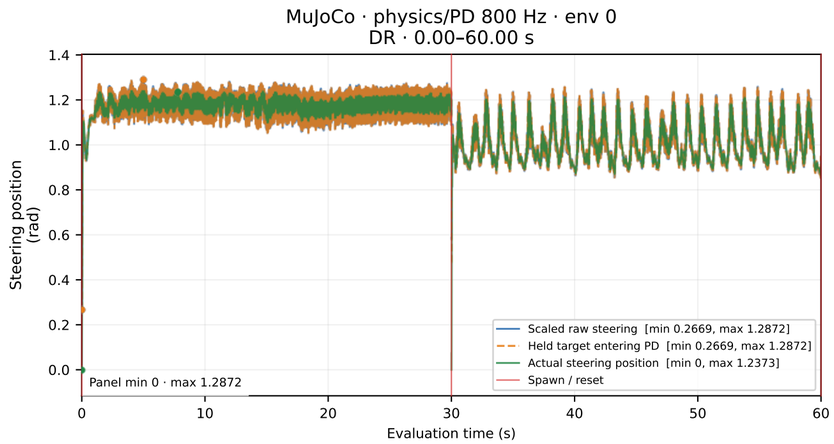
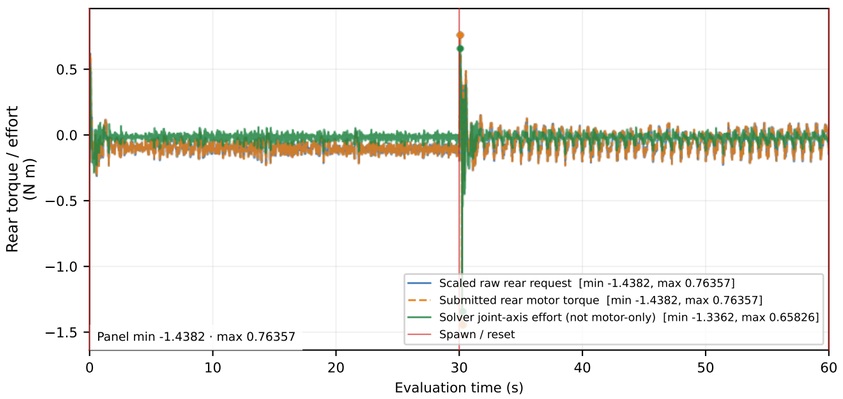
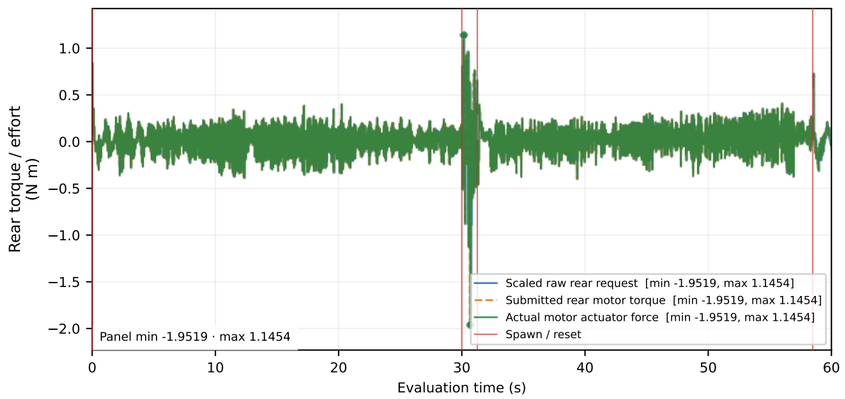
# model_2500 · Demand versus actuation · env 0 · full 60 s

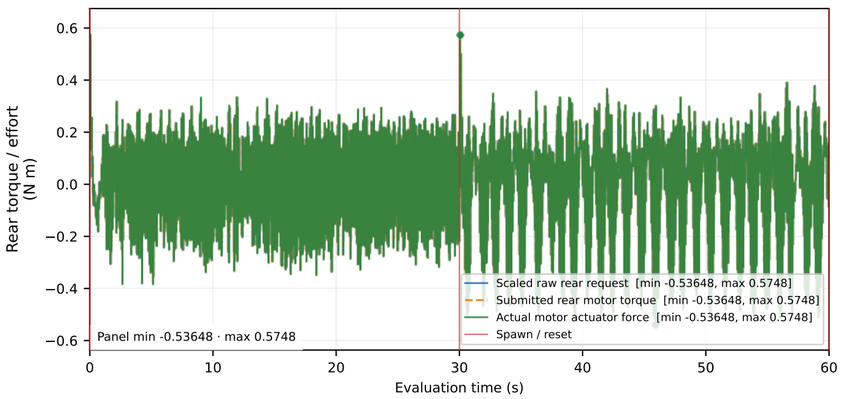


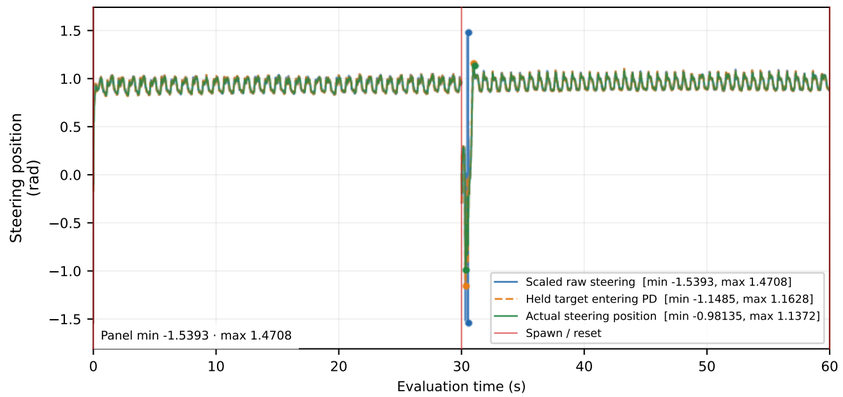

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

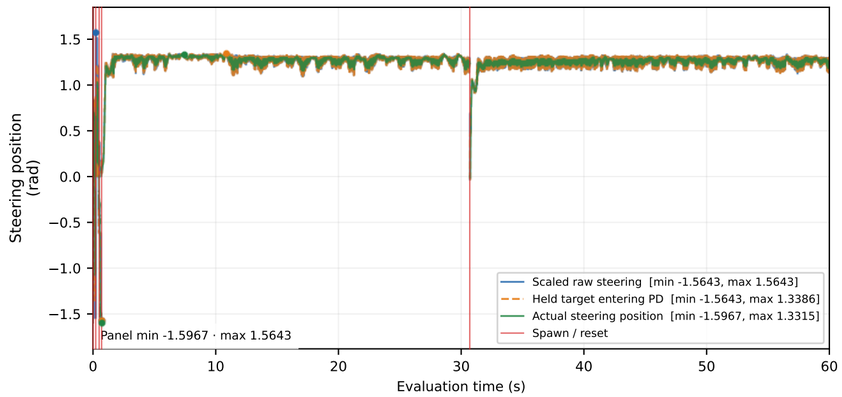
```
SAMPLED DR: env 0; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.04518, 0.804024, 0.959366, 0.816123]
  CoM XYZ mm [4.87718, -3.711, 0.621036]
  mu_s [0.741039, 0.681711, 0.592046, 0.532645]
  mu_d [0.614405, 0.611756, 0.534947, 0.532645]
  Kp [steer,rear] [10.6517, 0]; Kd [0.209184, 0]
  spawn roll 0.0198978 rad; rate 0.00125125 rad/s
  gyro bias XYZ rad/s [-0.000195088, -0.000150685, -2.64955e-05]
  gravity bias YZ [0.00115356, 0.00192213]
E2 [30.00, 60.00) s:
  mass x [1.07174, 1.18127, 1.11706, 0.881481]
  CoM XYZ mm [-0.792429, -4.35209, 3.72319]
  mu_s [0.576972, 0.659445, 0.531594, 0.547743]
  mu_d [0.576972, 0.547619, 0.531594, 0.547743]
  Kp [steer,rear] [9.66427, 0]; Kd [0.207538, 0]
  spawn roll -0.0845213 rad; rate 0.112094 rad/s
  gyro bias XYZ rad/s [-2.89794e-05, -3.58017e-05, 0.000157862]
  gravity bias YZ [-0.0029177, -0.00080667]
Six significant digits shown; full precision: sampled_dr_values.json.
```

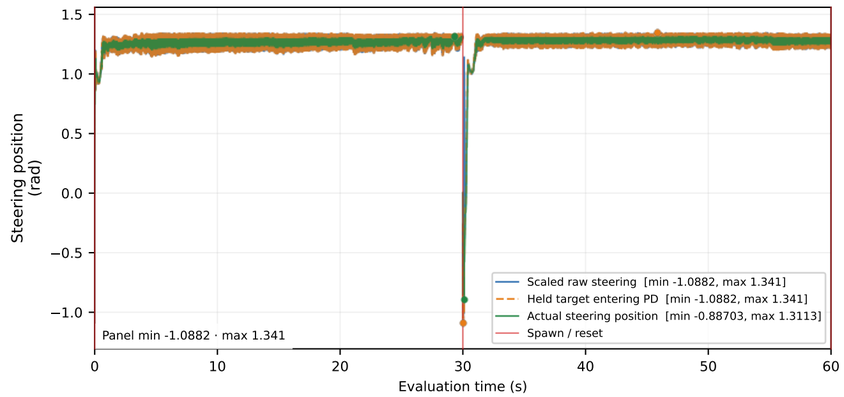
```
SAMPLED DR: env 0; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.05389, 1.02143, 1.02368, 0.92158]
  CoM XYZ mm [-4.69182, -0.632826, -2.85415]
  mu_s [0.666375, 0.640016, 0.669571, 0.71155]
  mu_d [0.519145, 0.513141, 0.669571, 0.71155]
  Kp [steer,rear] [9.46788, 0]; Kd [0.191255, 0]
  spawn roll 0.0628041 rad; rate -0.166514 rad/s
  gyro bias XYZ rad/s [0.000138097, -8.54505e-05, 3.17079e-05]
  gravity bias YZ [0.00125118, -0.000618693]
E2 [30.00, 31.26) s:
  mass x [1.02693, 1.08918, 0.965679, 0.927341]
  CoM XYZ mm [1.65679, 1.84224, -2.29571]
  mu_s [0.541939, 0.732187, 0.611239, 0.700521]
  mu_d [0.534359, 0.631664, 0.611239, 0.641329]
  Kp [steer,rear] [10.5106, 0]; Kd [0.202627, 0]
  spawn roll 0.059515 rad; rate -0.0147133 rad/s
  gyro bias XYZ rad/s [-4.78947e-06, 5.43307e-05, 0.000350425]
  gravity bias YZ [-0.000315206, -0.00188852]
E3 [31.26, 58.48) s:
  mass x [0.831615, 1.11129, 0.86509, 1.17322]
  CoM XYZ mm [4.90593, -3.7759, 1.83006]
  mu_s [0.669571, 0.546287, 0.69316, 0.640867]
  mu_d [0.669571, 0.546287, 0.69316, 0.556841]
  Kp [steer,rear] [9.35915, 0]; Kd [0.186665, 0]
  spawn roll -0.0194645 rad; rate -0.0189324 rad/s
  gyro bias XYZ rad/s [9.16669e-05, -2.25439e-05, 8.03725e-05]
  gravity bias YZ [0.00145273, 0.000686634]
E4 [58.48, 60.00) s:
  mass x [1.14858, 0.965632, 0.91743, 0.960669]
  CoM XYZ mm [-4.05657, 3.38234, 4.15215]
  mu_s [0.723429, 0.71008, 0.69316, 0.546287]
  mu_d [0.723429, 0.59371, 0.69316, 0.546287]
  Kp [steer,rear] [10.2283, 0]; Kd [0.182062, 0]
  spawn roll 0.085862 rad; rate 0.244715 rad/s
  gyro bias XYZ rad/s [-8.09711e-05, 0.000283962, 0.000268881]
  gravity bias YZ [-0.000732425, -0.000166156]
Six significant digits shown; full precision: sampled_dr_values.json.
```

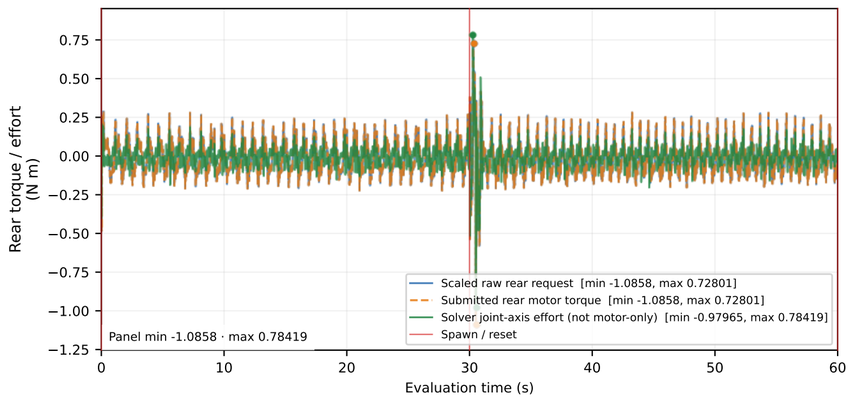
```
SAMPLED DR: env 0; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.05389, 1.02143, 1.02368, 0.92158]
  CoM XYZ mm [-4.69182, -0.632826, -2.85415]
  mu_s [0.666375, 0.640016, 0.669571, 0.71155]
  mu_d [0.519145, 0.513141, 0.669571, 0.71155]
  Kp [steer,rear] [9.46788, 0]; Kd [0.191255, 0]
  spawn roll 0.0628041 rad; rate -0.166514 rad/s
  gyro bias XYZ rad/s [0.000138097, -8.54505e-05, 3.17079e-05]
  gravity bias YZ [0.00125118, -0.000618693]
E2 [30.00, 60.00) s:
  mass x [1.02693, 1.08918, 0.965679, 0.927341]
  CoM XYZ mm [1.65679, 1.84224, -2.29571]
  mu_s [0.541939, 0.732187, 0.611239, 0.700521]
  mu_d [0.534359, 0.631664, 0.611239, 0.641329]
  Kp [steer,rear] [10.5106, 0]; Kd [0.202627, 0]
  spawn roll 0.059515 rad; rate -0.0147133 rad/s
  gyro bias XYZ rad/s [-4.78947e-06, 5.43307e-05, 0.000350425]
  gravity bias YZ [-0.000315206, -0.00188852]
Six significant digits shown; full precision: sampled_dr_values.json.
```

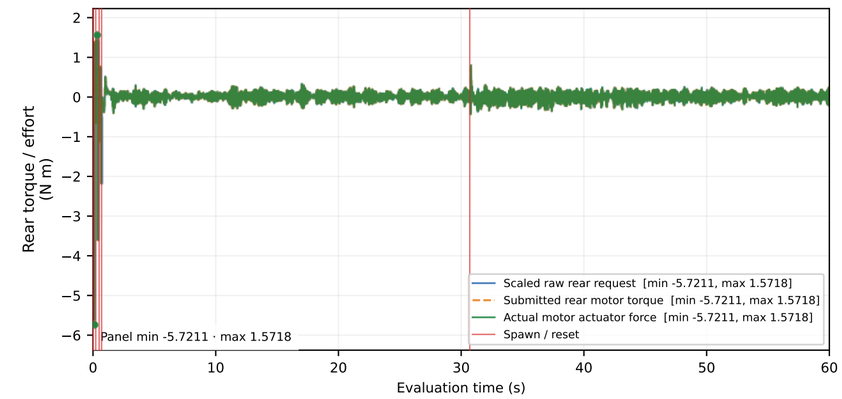
# model_2500 · Demand versus actuation · env 1 · full 60 s

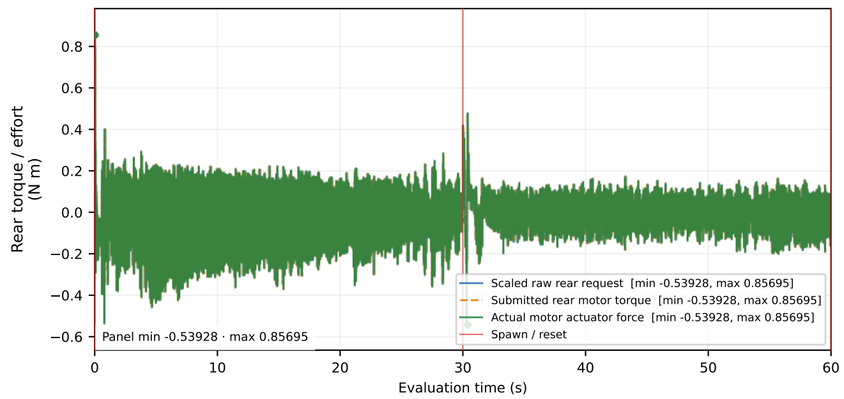


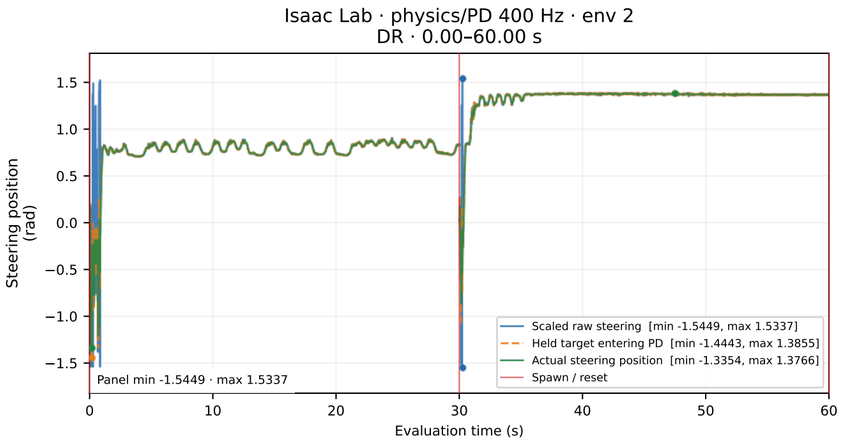

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

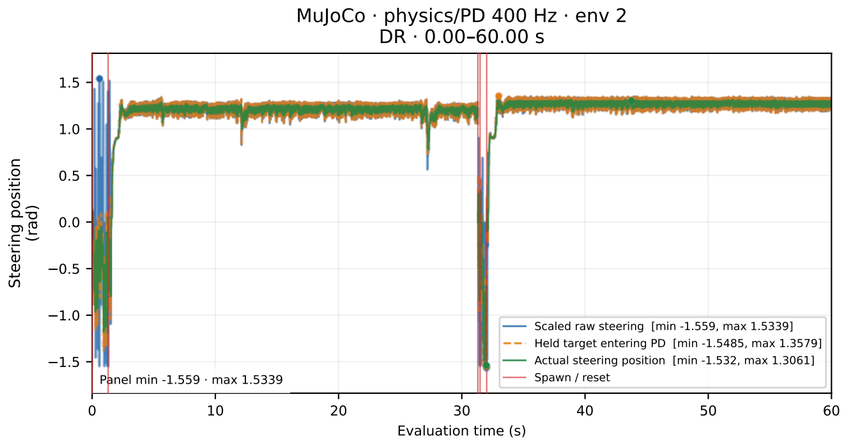
```
SAMPLED DR: env 1; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.862507, 0.992989, 1.0945, 0.962393]
  CoM XYZ mm [0.221074, 2.44478, 0.954762]
  mu_s [0.674963, 0.787792, 0.559809, 0.533455]
  mu_d [0.601294, 0.617134, 0.559809, 0.533455]
  Kp [steer,rear] [9.42558, 0]; Kd [0.184774, 0]
  spawn roll 0.0256746 rad; rate -0.215965 rad/s
  gyro bias XYZ rad/s [-7.03797e-05, 0.000323626, -7.54307e-05]
  gravity bias YZ [0.00224866, -0.00223355]
E2 [30.00, 60.00) s:
  mass x [1.17538, 0.984673, 0.876879, 1.09066]
  CoM XYZ mm [4.70234, 1.68571, 0.744864]
  mu_s [0.533455, 0.602694, 0.693205, 0.579974]
  mu_d [0.533455, 0.602694, 0.693205, 0.579974]
  Kp [steer,rear] [10.2626, 0]; Kd [0.183138, 0]
  spawn roll -0.00394535 rad; rate -0.198856 rad/s
  gyro bias XYZ rad/s [0.00023291, 2.05443e-07, 0.000381584]
  gravity bias YZ [-0.000380348, 0.00116294]
Six significant digits shown; full precision: sampled_dr_values.json.
```

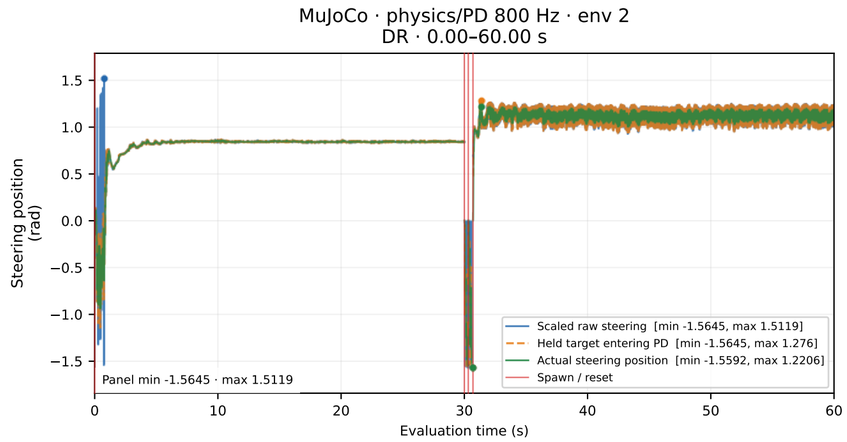
```
SAMPLED DR: env 1; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.22) s:
  mass x [1.06092, 0.81751, 0.808012, 1.13569]
  CoM XYZ mm [0.87143, -2.75295, 2.51792]
  mu_s [0.733515, 0.666375, 0.666375, 0.640016]
  mu_d [0.558392, 0.519145, 0.519145, 0.513141]
  Kp [steer,rear] [9.90206, 0]; Kd [0.215676, 0]
  spawn roll 0.067641 rad; rate 0.0175641 rad/s
  gyro bias XYZ rad/s [-7.13383e-05, -0.000159725, 5.31283e-06]
  gravity bias YZ [-0.00123234, 0.00117109]
E2 [0.22, 0.50) s:
  mass x [1.18392, 1.04402, 1.17318, 0.884353]
  CoM XYZ mm [3.00329, -3.53677, 1.20887]
  mu_s [0.748289, 0.559972, 0.757579, 0.611239]
  mu_d [0.689499, 0.502209, 0.70921, 0.611239]
  Kp [steer,rear] [9.1824, 0]; Kd [0.199162, 0]
  spawn roll 0.0977902 rad; rate -0.0536535 rad/s
  gyro bias XYZ rad/s [-0.000120112, 0.000267945, 2.74423e-05]
  gravity bias YZ [0.00344303, -0.00277117]
E3 [0.50, 0.70) s:
  mass x [0.840627, 1.16411, 1.09729, 0.896648]
  CoM XYZ mm [0.769351, 4.12939, -0.722516]
  mu_s [0.791209, 0.749678, 0.541939, 0.616244]
  mu_d [0.767936, 0.741429, 0.534359, 0.586498]
  Kp [steer,rear] [10.6528, 0]; Kd [0.205113, 0]
  spawn roll -0.0936273 rad; rate 0.000719158 rad/s
  gyro bias XYZ rad/s [0.000257614, -0.000169141, 9.57854e-05]
  gravity bias YZ [0.000545115, 0.00259499]
E4 [0.70, 30.70) s:
  mass x [0.82748, 1.12533, 1.14601, 0.998736]
  CoM XYZ mm [-2.38536, -2.57866, 2.20175]
  mu_s [0.546287, 0.732187, 0.631146, 0.631146]
  mu_d [0.546287, 0.631664, 0.631146, 0.631146]
  Kp [steer,rear] [9.89664, 0]; Kd [0.210172, 0]
  spawn roll -0.033323 rad; rate 0.0957727 rad/s
  gyro bias XYZ rad/s [9.53978e-05, 0.000048847, -0.000220352]
  gravity bias YZ [-0.000144735, 0.00422537]
E5 [30.70, 60.00) s:
  mass x [0.970729, 1.05628, 1.13447, 1.13934]
  CoM XYZ mm [-0.74075, -1.87991, 1.53457]
  mu_s [0.559972, 0.733515, 0.727426, 0.538976]
  mu_d [0.502209, 0.558392, 0.606358, 0.538976]
  Kp [steer,rear] [9.27399, 0]; Kd [0.20469, 0]
  spawn roll 0.043123 rad; rate -0.215016 rad/s
  gyro bias XYZ rad/s [-0.00054727, 0.000248029, -0.000156768]
  gravity bias YZ [-8.41541e-05, -0.00221894]
Six significant digits shown; full precision: sampled_dr_values.json.
```

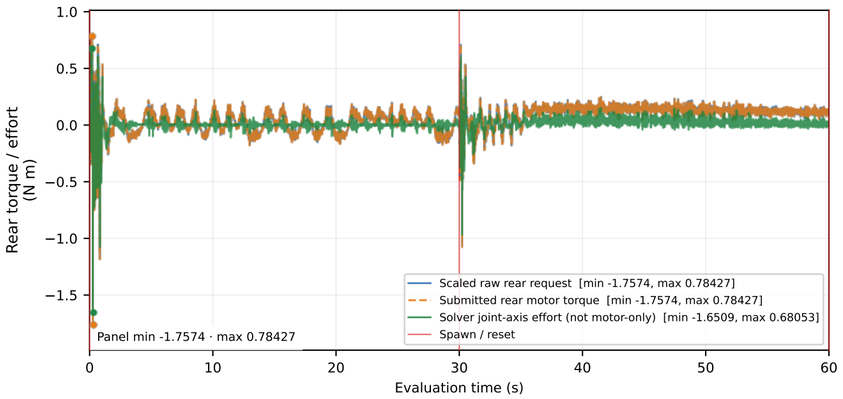
```
SAMPLED DR: env 1; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.06092, 0.81751, 0.808012, 1.13569]
  CoM XYZ mm [0.87143, -2.75295, 2.51792]
  mu_s [0.733515, 0.666375, 0.666375, 0.640016]
  mu_d [0.558392, 0.519145, 0.519145, 0.513141]
  Kp [steer,rear] [9.90206, 0]; Kd [0.215676, 0]
  spawn roll 0.067641 rad; rate 0.0175641 rad/s
  gyro bias XYZ rad/s [-7.13383e-05, -0.000159725, 5.31283e-06]
  gravity bias YZ [-0.00123234, 0.00117109]
E2 [30.00, 60.00) s:
  mass x [0.968901, 1.16355, 0.949248, 1.00463]
  CoM XYZ mm [-2.82922, -3.78885, 2.31511]
  mu_s [0.546287, 0.736077, 0.528253, 0.757579]
  mu_d [0.546287, 0.699455, 0.528253, 0.70921]
  Kp [steer,rear] [9.91541, 0]; Kd [0.217799, 0]
  spawn roll -0.0512449 rad; rate -0.0492599 rad/s
  gyro bias XYZ rad/s [4.82627e-05, -3.74329e-05, 0.000101373]
  gravity bias YZ [0.00136252, 0.00314652]
Six significant digits shown; full precision: sampled_dr_values.json.
```

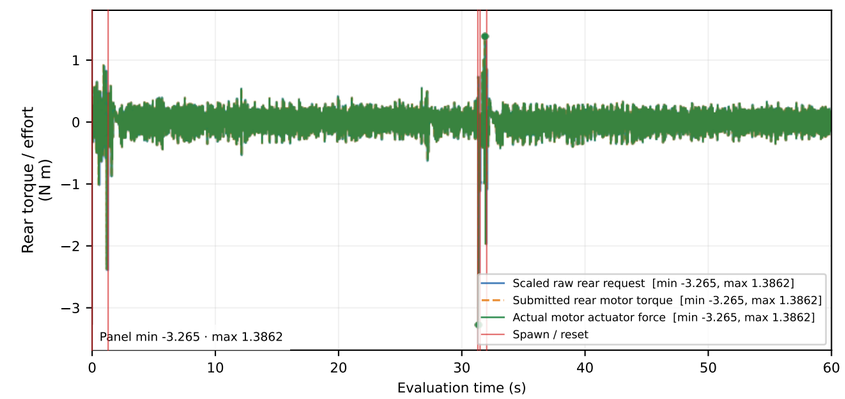
# model_2500 · Demand versus actuation · env 2 · full 60 s

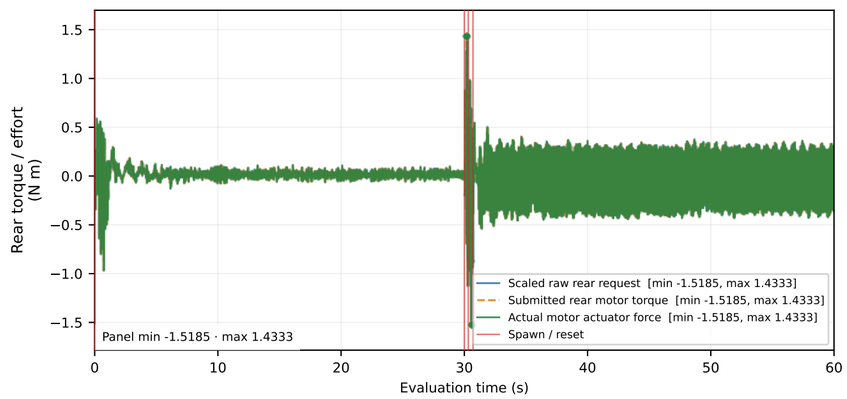


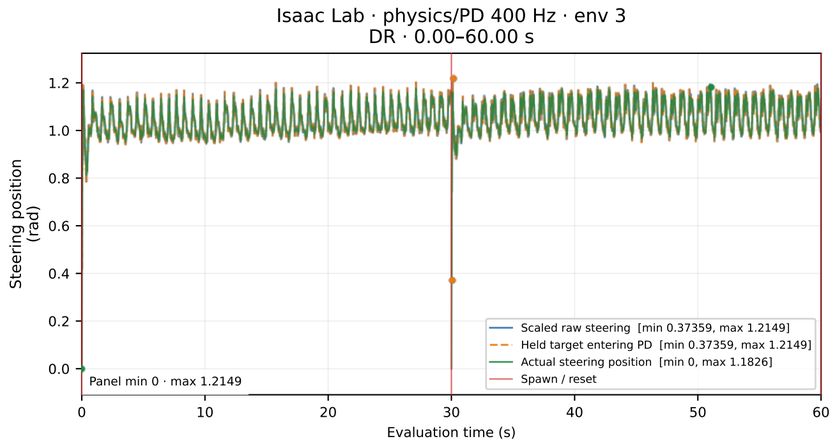

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

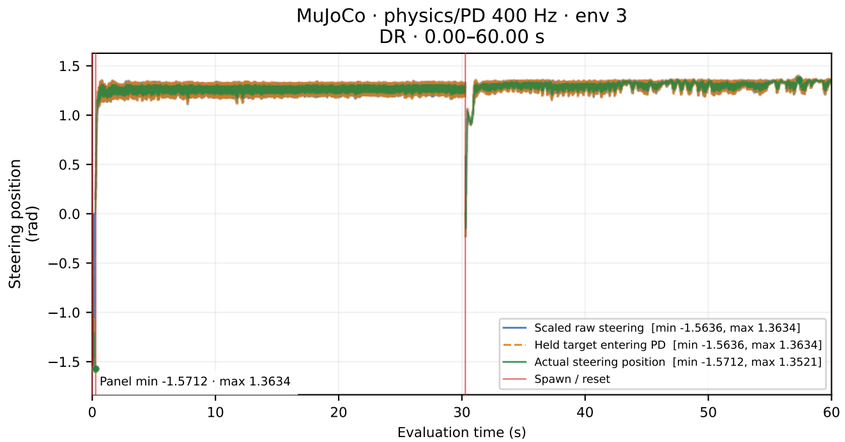
```
SAMPLED DR: env 2; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.00756, 0.914684, 0.896632, 1.16914]
  CoM XYZ mm [4.64708, 3.97917, 2.72967]
  mu_s [0.539956, 0.602694, 0.681711, 0.584424]
  mu_d [0.539956, 0.602694, 0.611756, 0.584424]
  Kp [steer,rear] [10.4584, 0]; Kd [0.182042, 0]
  spawn roll -0.0551139 rad; rate 0.115813 rad/s
  gyro bias XYZ rad/s [-0.000203379, -3.18315e-05, -0.000316606]
  gravity bias YZ [-0.00319071, -0.000254446]
E2 [30.00, 60.00) s:
  mass x [0.957834, 0.884986, 1.1579, 1.16429]
  CoM XYZ mm [-4.14227, -3.11448, -0.689477]
  mu_s [0.584424, 0.584424, 0.547743, 0.764681]
  mu_d [0.584424, 0.584424, 0.547743, 0.764681]
  Kp [steer,rear] [10.343, 0]; Kd [0.219287, 0]
  spawn roll -0.050929 rad; rate 0.070442 rad/s
  gyro bias XYZ rad/s [-0.000212198, -0.000398633, 0.000157541]
  gravity bias YZ [-0.00211123, 0.00306208]
Six significant digits shown; full precision: sampled_dr_values.json.
```

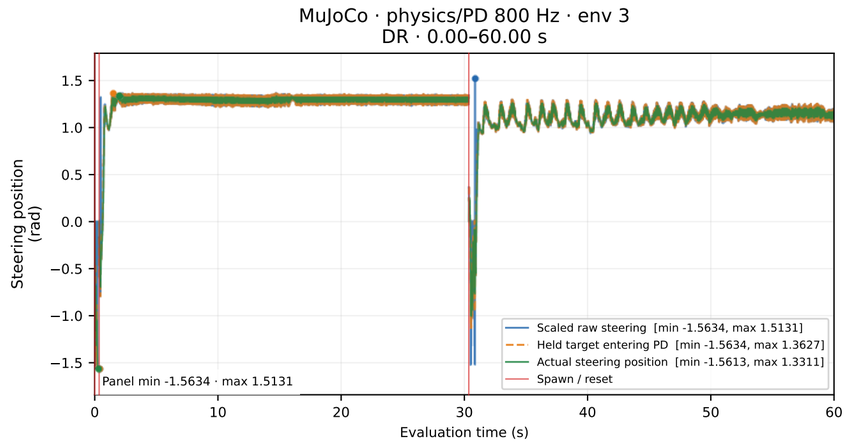
```
SAMPLED DR: env 2; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.30) s:
  mass x [0.849026, 0.903245, 0.962308, 1.18767]
  CoM XYZ mm [-3.37683, 3.57294, -3.36955]
  mu_s [0.728342, 0.727426, 0.733515, 0.71008]
  mu_d [0.728342, 0.606358, 0.558392, 0.59371]
  Kp [steer,rear] [10.2331, 0]; Kd [0.196456, 0]
  spawn roll 0.004772566 rad; rate -0.151419 rad/s
  gyro bias XYZ rad/s [5.35451e-05, 0.000141836, 3.5504e-05]
  gravity bias YZ [-0.00156514, 0.000524658]
E2 [1.30, 31.30) s:
  mass x [0.862338, 1.16887, 1.14804, 1.0074]
  CoM XYZ mm [-1.55679, -1.06343, -4.16265]
  mu_s [0.749678, 0.546287, 0.640867, 0.666375]
  mu_d [0.741429, 0.546287, 0.556841, 0.519145]
  Kp [steer,rear] [9.15738, 0]; Kd [0.211983, 0]
  spawn roll -0.045117 rad; rate 0.104407 rad/s
  gyro bias XYZ rad/s [0.000274867, -4.48659e-05, -0.000265008]
  gravity bias YZ [0.00171783, -0.000824402]
E3 [31.30, 31.48) s:
  mass x [0.955405, 1.09469, 0.992401, 0.887005]
  CoM XYZ mm [-4.13751, -3.91435, 4.85292]
  mu_s [0.633024, 0.546287, 0.640867, 0.732187]
  mu_d [0.568172, 0.546287, 0.556841, 0.631664]
  Kp [steer,rear] [9.8247, 0]; Kd [0.218829, 0]
  spawn roll 0.0975943 rad; rate 0.188861 rad/s
  gyro bias XYZ rad/s [-5.2756e-06, 0.000223486, -5.41635e-05]
  gravity bias YZ [0.00141839, -0.00176458]
E4 [31.48, 32.02) s:
  mass x [0.822835, 0.939556, 1.09824, 0.922163]
  CoM XYZ mm [1.99899, 3.27979, 2.17474]
  mu_s [0.791209, 0.727426, 0.669571, 0.637675]
  mu_d [0.767936, 0.606358, 0.669571, 0.637675]
  Kp [steer,rear] [10.7052, 0]; Kd [0.183295, 0]
  spawn roll -0.00791113 rad; rate 0.138752 rad/s
  gyro bias XYZ rad/s [-0.000304359, 0.000170522, -0.000105578]
  gravity bias YZ [0.00262822, 0.00585244]
E5 [32.02, 60.00) s:
  mass x [0.866334, 0.98371, 1.17728, 1.17219]
  CoM XYZ mm [1.371, -2.09598, -1.37734]
  mu_s [0.541939, 0.633024, 0.538976, 0.597748]
  mu_d [0.534359, 0.568172, 0.538976, 0.597748]
  Kp [steer,rear] [9.16109, 0]; Kd [0.199155, 0]
  spawn roll -0.00517896 rad; rate -0.125551 rad/s
  gyro bias XYZ rad/s [-0.000215667, 4.98964e-05, -0.000202771]
  gravity bias YZ [0.00045478, -0.00260641]
Six significant digits shown; full precision: sampled_dr_values.json.
```

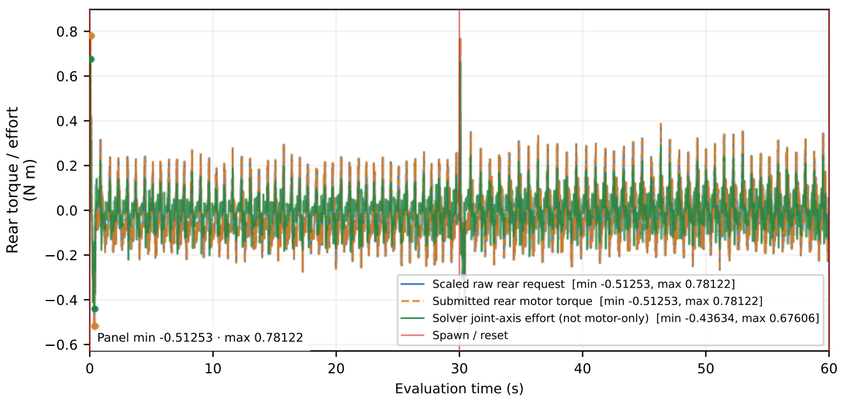
```
SAMPLED DR: env 2; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.849026, 0.903245, 0.962308, 1.18767]
  CoM XYZ mm [-3.37683, 3.57294, -3.36955]
  mu_s [0.728342, 0.727426, 0.733515, 0.71008]
  mu_d [0.728342, 0.606358, 0.558392, 0.59371]
  Kp [steer,rear] [10.2331, 0]; Kd [0.196456, 0]
  spawn roll 0.00477256 rad; rate -0.151419 rad/s
  gyro bias XYZ rad/s [5.35451e-05, 0.000141836, 3.5504e-05]
  gravity bias YZ [-0.00156514, 0.000524658]
E2 [30.00, 30.32) s:
  mass x [1.15833, 0.950965, 1.07119, 0.986383]
  CoM XYZ mm [-2.28574, 4.94326, 0.580657]
  mu_s [0.538976, 0.541939, 0.700521, 0.791209]
  mu_d [0.538976, 0.534359, 0.641329, 0.767936]
  Kp [steer,rear] [9.96591, 0]; Kd [0.207815, 0]
  spawn roll -0.0830041 rad; rate 0.109036 rad/s
  gyro bias XYZ rad/s [-8.98043e-05, 5.78866e-05, 0.000250741]
  gravity bias YZ [-0.000604908, -0.00123248]
E3 [30.32, 30.70) s:
  mass x [0.832338, 0.882538, 1.0699, 1.17094]
  CoM XYZ mm [2.00383, 4.7638, 1.17088]
  mu_s [0.538434, 0.559972, 0.733515, 0.700521]
  mu_d [0.538434, 0.502209, 0.558392, 0.641329]
  Kp [steer,rear] [10.6028, 0]; Kd [0.20534, 0]
  spawn roll -0.0550542 rad; rate 0.058826 rad/s
  gyro bias XYZ rad/s [-1.2178e-05, 0.000148104, 0.000107851]
  gravity bias YZ [-0.00203786, 0.00101602]
E4 [30.70, 60.00) s:
  mass x [1.04999, 0.922194, 1.06322, 0.970641]
  CoM XYZ mm [4.90851, 1.67046, -3.5585]
  mu_s [0.640016, 0.666375, 0.700521, 0.728342]
  mu_d [0.513141, 0.519145, 0.641329, 0.728342]
  Kp [steer,rear] [10.797, 0]; Kd [0.215121, 0]
  spawn roll 0.0438033 rad; rate 0.105346 rad/s
  gyro bias XYZ rad/s [0.000270291, -0.000103782, 2.09693e-05]
  gravity bias YZ [0.0017364, -0.00118816]
Six significant digits shown; full precision: sampled_dr_values.json.
```

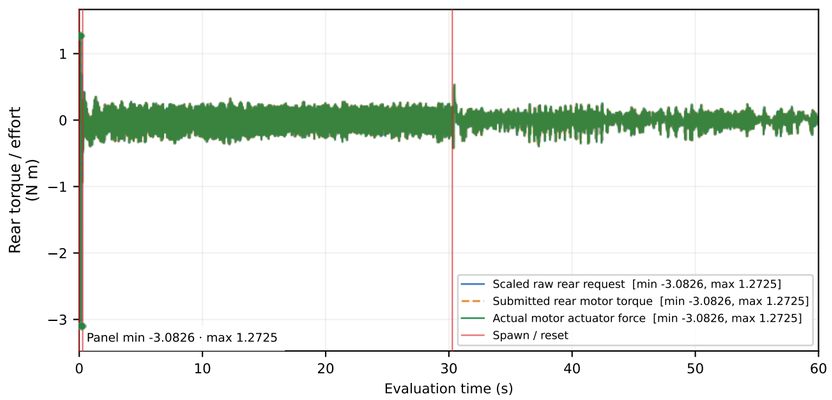
# model_2500 · Demand versus actuation · env 3 · full 60 s

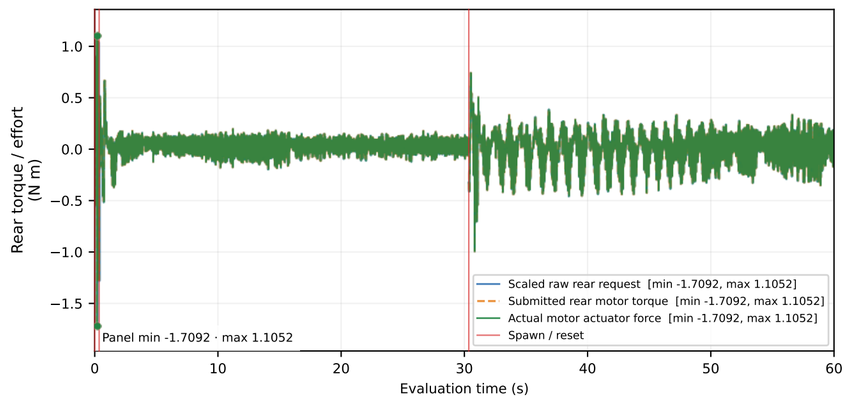


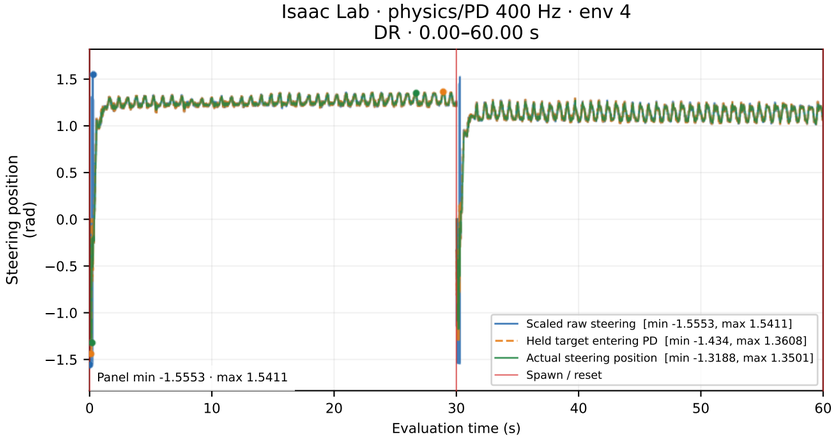

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

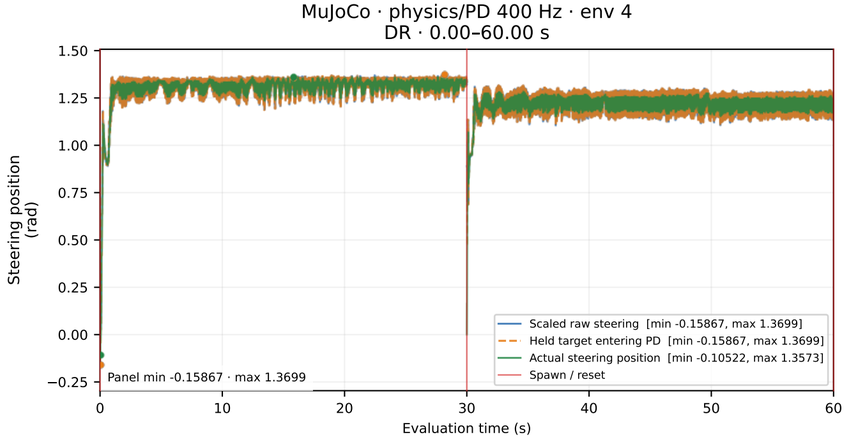
```
SAMPLED DR: env 3; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.13197, 0.813675, 0.955158, 0.832956]
  CoM XYZ mm [1.6811, 0.46216, 0.0711083]
  mu_s [0.713129, 0.632409, 0.602694, 0.774409]
  mu_d [0.713129, 0.589076, 0.602694, 0.561095]
  Kp [steer,rear] [9.6419, 0]; Kd [0.189538, 0]
  spawn roll 0.0698016 rad; rate -0.142062 rad/s
  gyro bias XYZ rad/s [0.000115749, -3.6422e-05, -0.000145327]
  gravity bias YZ [-0.00140168, -0.00330635]
E2 [30.00, 60.00) s:
  mass x [1.01015, 1.06654, 0.832539, 0.908403]
  CoM XYZ mm [-1.31282, 0.536375, 0.871569]
  mu_s [0.539956, 0.539956, 0.579974, 0.632409]
  mu_d [0.539956, 0.539956, 0.579974, 0.589076]
  Kp [steer,rear] [10.686, 0]; Kd [0.201728, 0]
  spawn roll 0.0615653 rad; rate -0.131084 rad/s
  gyro bias XYZ rad/s [-8.79698e-05, -0.000189956, -1.0838e-05]
  gravity bias YZ [-0.00019208, 0.000399016]
Six significant digits shown; full precision: sampled_dr_values.json.
```

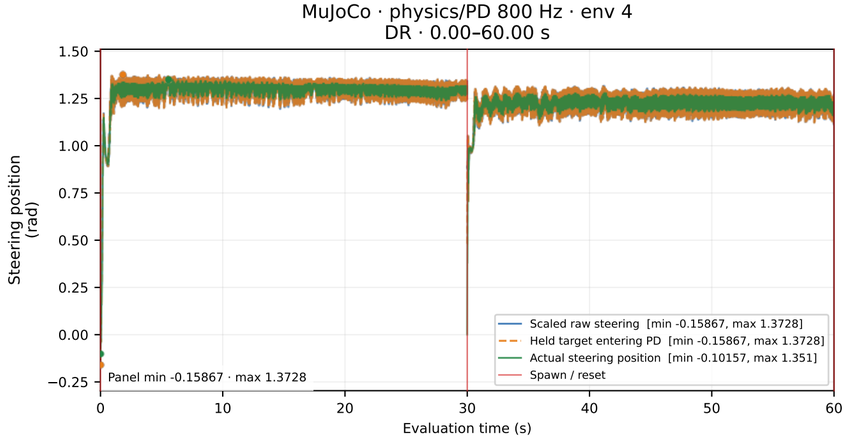
```
SAMPLED DR: env 3; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.28) s:
  mass x [1.02925, 1.0114, 1.10546, 1.12468]
  CoM XYZ mm [0.102289, 2.7934, 2.96113]
  mu_s [0.704749, 0.568073, 0.748289, 0.640016]
  mu_d [0.541926, 0.568073, 0.689499, 0.513141]
  Kp [steer,rear] [10.3426, 0]; Kd [0.213603, 0]
  spawn roll -0.0890225 rad; rate -0.19006 rad/s
  gyro bias XYZ rad/s [-4.54431e-05, -0.000198711, -0.000291781]
  gravity bias YZ [0.000127515, 0.00154904]
E2 [0.28, 30.28) s:
  mass x [0.837542, 1.15518, 0.926383, 1.19918]
  CoM XYZ mm [3.94932, -2.13762, 1.29165]
  mu_s [0.546287, 0.541939, 0.538434, 0.732187]
  mu_d [0.546287, 0.534359, 0.538434, 0.631664]
  Kp [steer,rear] [10.1341, 0]; Kd [0.183532, 0]
  spawn roll -0.00720207 rad; rate -0.000312354 rad/s
  gyro bias XYZ rad/s [2.58975e-05, -0.000105769, 0.000105602]
  gravity bias YZ [-0.00163113, -0.00077671]
E3 [30.28, 60.00) s:
  mass x [1.05434, 1.13786, 0.975343, 1.1978]
  CoM XYZ mm [-0.350068, -3.69408, -1.36237]
  mu_s [0.541939, 0.597748, 0.597748, 0.640867]
  mu_d [0.534359, 0.597748, 0.597748, 0.556841]
  Kp [steer,rear] [9.80054, 0]; Kd [0.190214, 0]
  spawn roll -0.000669958 rad; rate -0.187088 rad/s
  gyro bias XYZ rad/s [-3.15961e-05, 5.02063e-05, -0.000199858]
  gravity bias YZ [-0.00370918, 0.00131895]
Six significant digits shown; full precision: sampled_dr_values.json.
```

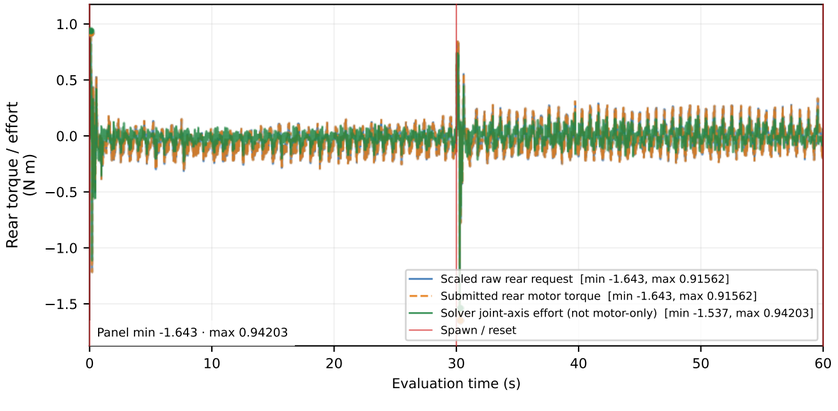
```
SAMPLED DR: env 3; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.36) s:
  mass x [1.02925, 1.0114, 1.10546, 1.12468]
  CoM XYZ mm [0.102289, 2.7934, 2.96113]
  mu_s [0.704749, 0.568073, 0.748289, 0.640016]
  mu_d [0.541926, 0.568073, 0.689499, 0.513141]
  Kp [steer,rear] [10.3426, 0]; Kd [0.213603, 0]
  spawn roll -0.0890225 rad; rate -0.19006 rad/s
  gyro bias XYZ rad/s [-4.54431e-05, -0.000198711, -0.000291781]
  gravity bias YZ [0.000127515, 0.00154904]
E2 [0.36, 30.36) s:
  mass x [1.19907, 1.06352, 1.14454, 1.04925]
  CoM XYZ mm [-0.034423, -4.44887, -0.841293]
  mu_s [0.748289, 0.546287, 0.640867, 0.732187]
  mu_d [0.689499, 0.546287, 0.556841, 0.631664]
  Kp [steer,rear] [10.6007, 0]; Kd [0.21073, 0]
  spawn roll -0.0493498 rad; rate -0.0192491 rad/s
  gyro bias XYZ rad/s [-8.97478e-05, -4.88573e-05, 0.000290481]
  gravity bias YZ [-0.000123604, 0.00476323]
E3 [30.36, 60.00) s:
  mass x [1.04058, 0.989516, 1.13018, 1.10386]
  CoM XYZ mm [-3.10637, 0.0190005, -0.501031]
  mu_s [0.71155, 0.723429, 0.728342, 0.597748]
  mu_d [0.71155, 0.723429, 0.728342, 0.597748]
  Kp [steer,rear] [10.6943, 0]; Kd [0.182485, 0]
  spawn roll -0.0463704 rad; rate 0.205555 rad/s
  gyro bias XYZ rad/s [0.000370802, 0.000462692, -0.000154158]
  gravity bias YZ [0.000957546, -0.00018204]
Six significant digits shown; full precision: sampled_dr_values.json.
```

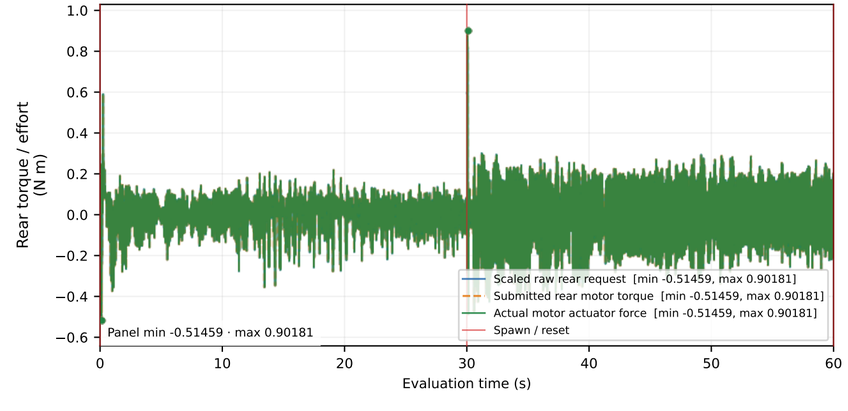
# model_2500 · Demand versus actuation · env 4 · full 60 s

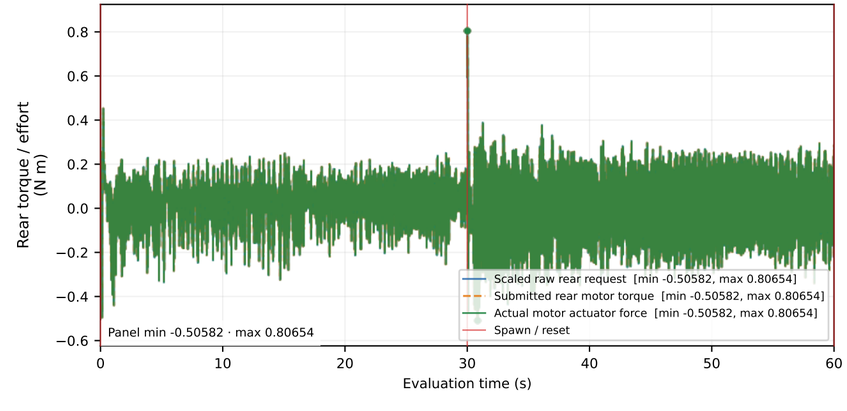


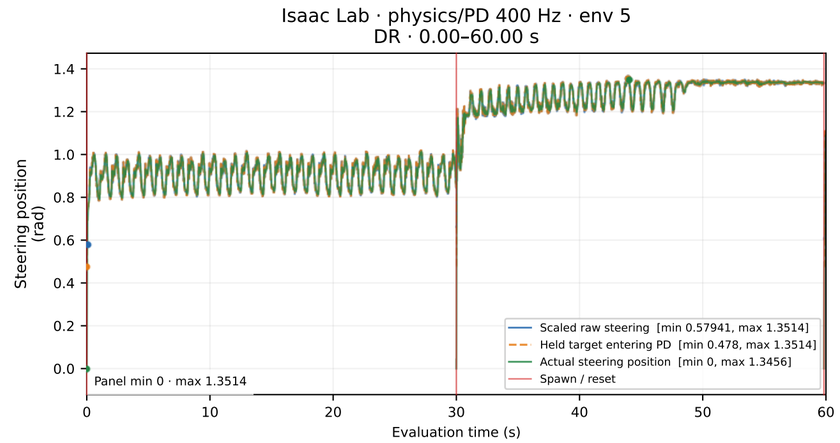

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

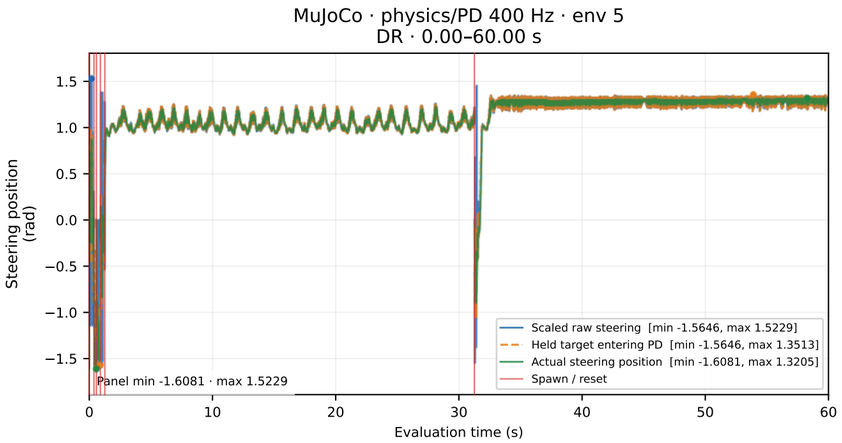
```
SAMPLED DR: env 4; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.10967, 0.911688, 1.00553, 0.882728]
  CoM XYZ mm [0.882208, -2.44468, 3.96283]
  mu_s [0.547743, 0.794338, 0.533455, 0.632409]
  mu_d [0.547743, 0.526209, 0.533455, 0.589076]
  Kp [steer,rear] [10.51, 0]; Kd [0.18806, 0]
  spawn roll -0.0996169 rad; rate -0.165478 rad/s
  gyro bias XYZ rad/s [-0.00025822, 9.37975e-06, 0.000278995]
  gravity bias YZ [0.000311097, 0.00214254]
E2 [30.00, 60.00) s:
  mass x [1.14475, 0.888204, 0.814729, 0.899869]
  CoM XYZ mm [0.76364, -0.353117, -3.2285]
  mu_s [0.726175, 0.760821, 0.579974, 0.547743]
  mu_d [0.558574, 0.670315, 0.579974, 0.547743]
  Kp [steer,rear] [10.4888, 0]; Kd [0.188838, 0]
  spawn roll -0.0724018 rad; rate -0.0422324 rad/s
  gyro bias XYZ rad/s [0.000232035, 0.000363317, -1.80533e-05]
  gravity bias YZ [-0.00186627, -0.000895573]
Six significant digits shown; full precision: sampled_dr_values.json.
```

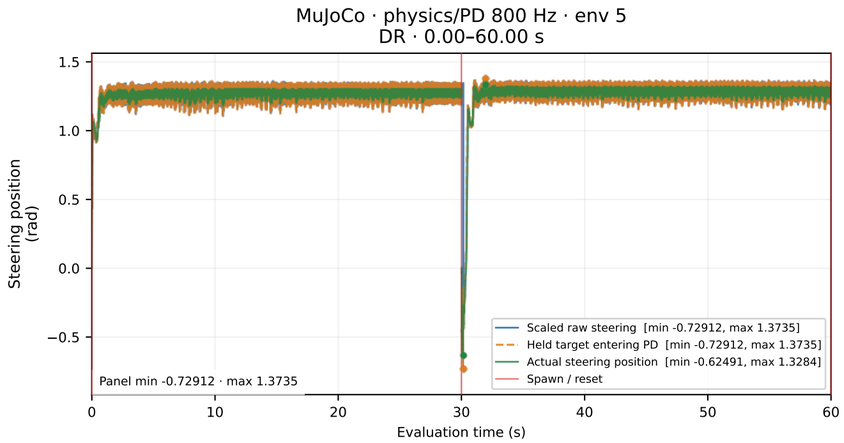
```
SAMPLED DR: env 4; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.16224, 0.830891, 0.909028, 1.04874]
  CoM XYZ mm [4.34028, -4.15053, 1.2923]
  mu_s [0.559972, 0.704749, 0.633024, 0.597748]
  mu_d [0.502209, 0.541926, 0.568172, 0.597748]
  Kp [steer,rear] [10.497, 0]; Kd [0.199966, 0]
  spawn roll -0.0219811 rad; rate -0.0869557 rad/s
  gyro bias XYZ rad/s [0.000105059, 6.3236e-05, -7.15947e-05]
  gravity bias YZ [-0.00185749, -0.00210565]
E2 [30.00, 60.00) s:
  mass x [0.818194, 1.13428, 0.985815, 1.09866]
  CoM XYZ mm [-0.129074, -1.07891, 2.27407]
  mu_s [0.559972, 0.757579, 0.541939, 0.568073]
  mu_d [0.502209, 0.70921, 0.534359, 0.568073]
  Kp [steer,rear] [10.0033, 0]; Kd [0.181016, 0]
  spawn roll 0.0661497 rad; rate 0.227347 rad/s
  gyro bias XYZ rad/s [5.09085e-05, 0.00023236, 7.71043e-05]
  gravity bias YZ [7.66479e-05, 0.00122516]
Six significant digits shown; full precision: sampled_dr_values.json.
```

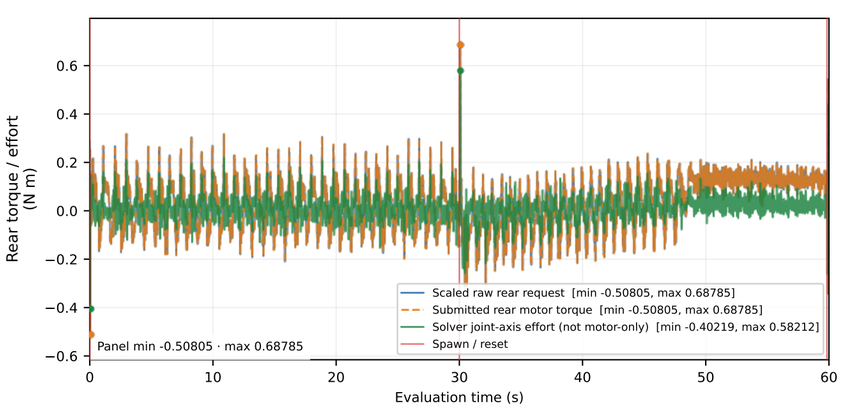
```
SAMPLED DR: env 4; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.16224, 0.830891, 0.909028, 1.04874]
  CoM XYZ mm [4.34028, -4.15053, 1.2923]
  mu_s [0.559972, 0.704749, 0.633024, 0.597748]
  mu_d [0.502209, 0.541926, 0.568172, 0.597748]
  Kp [steer,rear] [10.497, 0]; Kd [0.199966, 0]
  spawn roll -0.0219811 rad; rate -0.0869557 rad/s
  gyro bias XYZ rad/s [0.000105059, 6.3236e-05, -7.15947e-05]
  gravity bias YZ [-0.00185749, -0.00210565]
E2 [30.00, 60.00) s:
  mass x [0.818194, 1.13428, 0.985815, 1.09866]
  CoM XYZ mm [-0.129074, -1.07891, 2.27407]
  mu_s [0.559972, 0.757579, 0.541939, 0.568073]
  mu_d [0.502209, 0.70921, 0.534359, 0.568073]
  Kp [steer,rear] [10.0033, 0]; Kd [0.181016, 0]
  spawn roll 0.0661497 rad; rate 0.227347 rad/s
  gyro bias XYZ rad/s [5.09085e-05, 0.00023236, 7.71043e-05]
  gravity bias YZ [7.66479e-05, 0.00122516]
Six significant digits shown; full precision: sampled_dr_values.json.
```

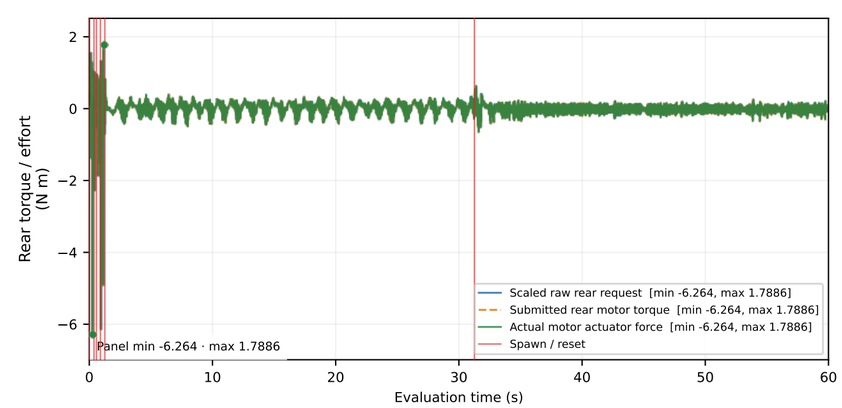
# model_2500 · Demand versus actuation · env 5 · full 60 s

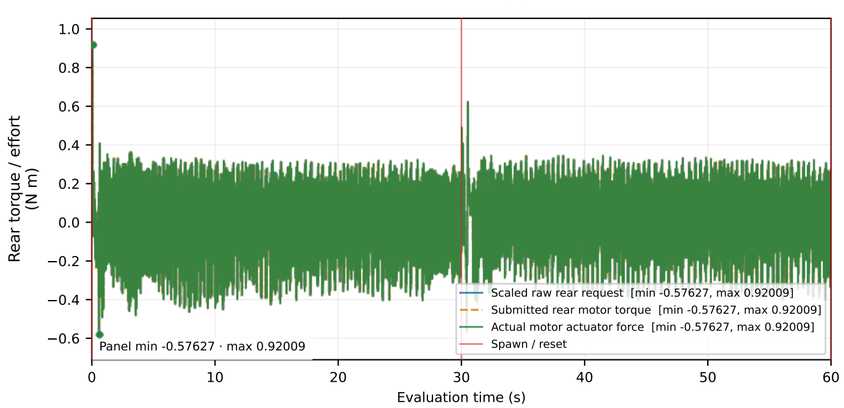


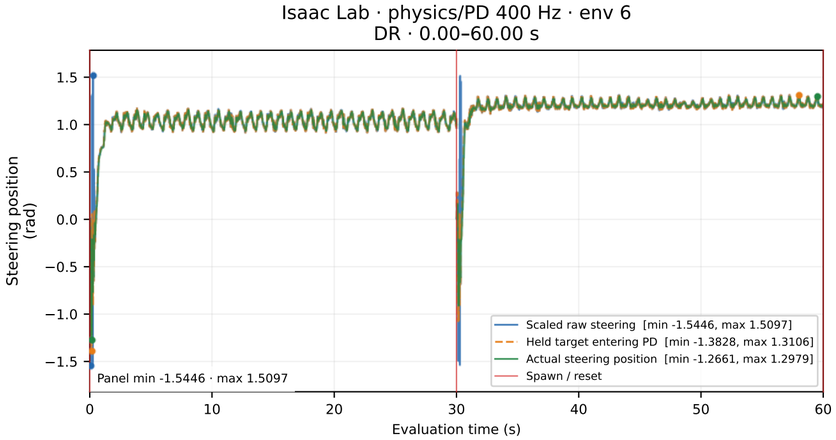

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

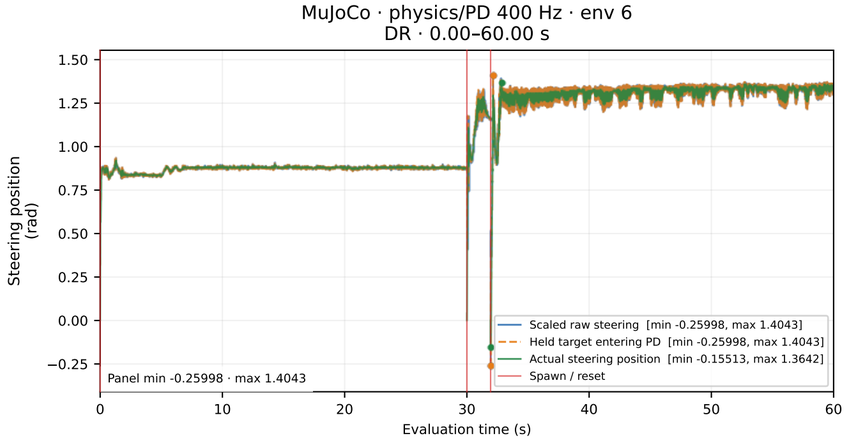
```
SAMPLED DR: env 5; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.802947, 1.15001, 1.18729, 0.848047]
  CoM XYZ mm [-3.13438, 2.23542, -3.2187]
  mu_s [0.659445, 0.584424, 0.673378, 0.592046]
  mu_d [0.547619, 0.584424, 0.673378, 0.534947]
  Kp [steer,rear] [9.42645, 0]; Kd [0.205456, 0]
  spawn roll 0.0161676 rad; rate 0.204907 rad/s
  gyro bias XYZ rad/s [0.000204662, 0.000224783, -8.47435e-05]
  gravity bias YZ [-0.000148007, -0.00267725]
E2 [30.00, 59.84) s:
  mass x [0.881139, 1.13225, 0.949404, 1.08537]
  CoM XYZ mm [0.0505447, -1.46302, 3.60368]
  mu_s [0.647391, 0.741039, 0.647391, 0.539956]
  mu_d [0.647391, 0.614405, 0.647391, 0.539956]
  Kp [steer,rear] [9.62694, 0]; Kd [0.189018, 0]
  spawn roll 0.0462433 rad; rate 0.0846739 rad/s
  gyro bias XYZ rad/s [9.06762e-05, 0.000140041, -6.86844e-05]
  gravity bias YZ [0.00423396, -0.000195554]
E3 [59.84, 60.00) s:
  mass x [1.12477, 1.18441, 0.806715, 0.806501]
  CoM XYZ mm [-2.09889, -3.31393, -0.832587]
  mu_s [0.703711, 0.592046, 0.681711, 0.713129]
  mu_d [0.703711, 0.534947, 0.611756, 0.713129]
  Kp [steer,rear] [9.37835, 0]; Kd [0.192918, 0]
  spawn roll -0.000824928 rad; rate 0.175743 rad/s
  gyro bias XYZ rad/s [7.24127e-05, -0.000395092, 0.000189631]
  gravity bias YZ [-0.00136036, -0.00204384]
Six significant digits shown; full precision: sampled_dr_values.json.
```

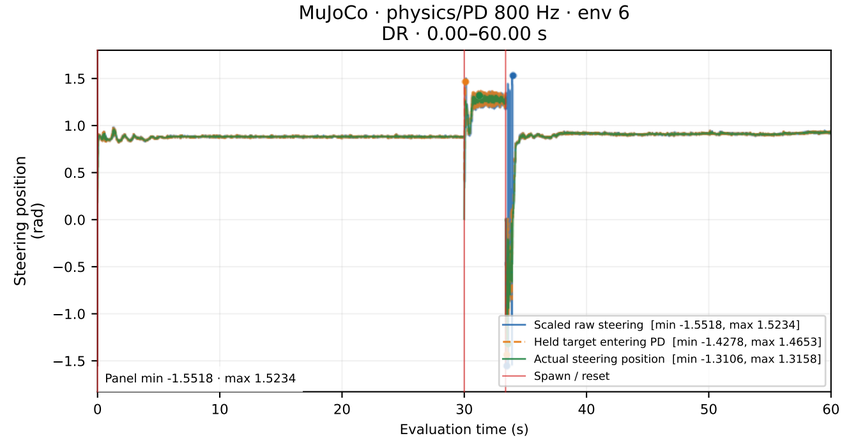
```
SAMPLED DR: env 5; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.38) s:
  mass x [1.09672, 1.10147, 0.986072, 0.84149]
  CoM XYZ mm [4.66852, -1.7925, -2.99668]
  mu_s [0.640016, 0.71155, 0.538434, 0.597748]
  mu_d [0.513141, 0.71155, 0.538434, 0.597748]
  Kp [steer,rear] [9.35848, 0]; Kd [0.211719, 0]
  spawn roll 0.0804349 rad; rate -0.153191 rad/s
  gyro bias XYZ rad/s [0.000176228, 0.000366092, -0.000125907]
  gravity bias YZ [0.00320495, -0.00211962]
E2 [0.38, 0.58) s:
  mass x [1.11491, 1.15873, 0.825652, 0.899697]
  CoM XYZ mm [4.36758, 0.878995, -4.68893]
  mu_s [0.538434, 0.616244, 0.611239, 0.728342]
  mu_d [0.538434, 0.586498, 0.611239, 0.728342]
  Kp [steer,rear] [10.8842, 0]; Kd [0.210061, 0]
  spawn roll -0.0855557 rad; rate -0.247262 rad/s
  gyro bias XYZ rad/s [-0.000168705, -8.27003e-05, 7.35071e-05]
  gravity bias YZ [7.95918e-05, -0.00316247]
E3 [0.58, 0.90) s:
  mass x [0.966119, 0.831289, 1.03341, 1.09674]
  CoM XYZ mm [-0.0965522, 4.67369, -4.64946]
  mu_s [0.727426, 0.538434, 0.538434, 0.528253]
  mu_d [0.606358, 0.538434, 0.538434, 0.528253]
  Kp [steer,rear] [10.4336, 0]; Kd [0.186615, 0]
  spawn roll -0.0877659 rad; rate -0.00385789 rad/s
  gyro bias XYZ rad/s [0.000231287, 0.000186892, 0.000189582]
  gravity bias YZ [-0.00284522, 0.00157655]
E4 [0.90, 1.26) s:
  mass x [0.978842, 1.15451, 1.09598, 0.821619]
  CoM XYZ mm [1.16774, -4.98495, 0.0479427]
  mu_s [0.791209, 0.616244, 0.71155, 0.568073]
  mu_d [0.767936, 0.586498, 0.71155, 0.568073]
  Kp [steer,rear] [10.9289, 0]; Kd [0.209175, 0]
  spawn roll -0.0261108 rad; rate -0.102054 rad/s
  gyro bias XYZ rad/s [-6.47184e-05, 3.55993e-05, -0.000303179]
  gravity bias YZ [0.00180537, 2.05281e-05]
E5 [1.26, 31.26) s:
  mass x [1.11897, 1.17619, 1.12289, 1.14132]
  CoM XYZ mm [-0.575743, 1.12483, -2.3802]
  mu_s [0.541939, 0.71008, 0.71008, 0.733515]
  mu_d [0.534359, 0.59371, 0.59371, 0.558392]
  Kp [steer,rear] [10.6454, 0]; Kd [0.213398, 0]
  spawn roll 0.0508153 rad; rate 0.225338 rad/s
  gyro bias XYZ rad/s [1.53426e-05, -0.000113271, -6.7393e-05]
  gravity bias YZ [0.0019273, 0.00144469]
E6 [31.26, 60.00) s:
  mass x [1.08294, 0.856463, 0.828188, 1.19957]
  CoM XYZ mm [1.13228, -3.49657, 3.12181]
  mu_s [0.631146, 0.616244, 0.568073, 0.728342]
  mu_d [0.631146, 0.586498, 0.568073, 0.728342]
  Kp [steer,rear] [9.07231, 0]; Kd [0.212301, 0]
  spawn roll -0.0509221 rad; rate -0.230631 rad/s
  gyro bias XYZ rad/s [2.57357e-05, -0.000184562, -0.000389171]
  gravity bias YZ [-0.00143281, -0.00148157]
Six significant digits shown; full precision: sampled_dr_values.json.
```

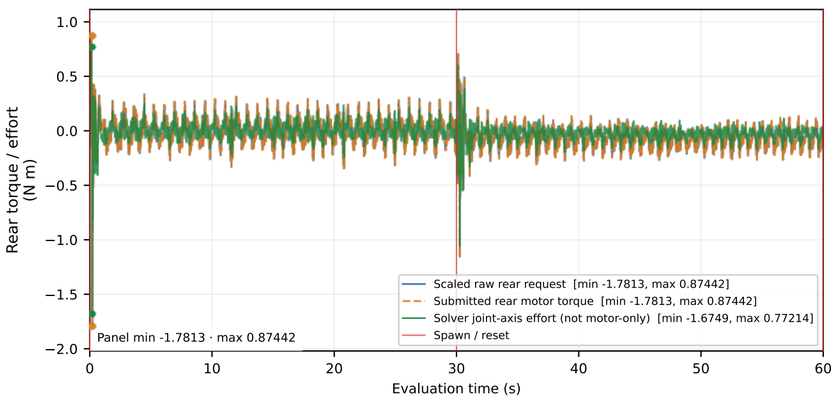
```
SAMPLED DR: env 5; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09672, 1.10147, 0.986072, 0.84149]
  CoM XYZ mm [4.66852, -1.7925, -2.99668]
  mu_s [0.640016, 0.71155, 0.538434, 0.597748]
  mu_d [0.513141, 0.71155, 0.538434, 0.597748]
  Kp [steer,rear] [9.35848, 0]; Kd [0.211719, 0]
  spawn roll 0.0804349 rad; rate -0.153191 rad/s
  gyro bias XYZ rad/s [0.000176228, 0.000366092, -0.000125907]
  gravity bias YZ [0.00320495, -0.00211962]
E2 [30.00, 60.00) s:
  mass x [0.997624, 1.11207, 0.952525, 0.89016]
  CoM XYZ mm [2.79459, -2.94807, -4.33668]
  mu_s [0.640867, 0.666375, 0.700521, 0.616244]
  mu_d [0.556841, 0.519145, 0.641329, 0.586498]
  Kp [steer,rear] [10.059, 0]; Kd [0.186785, 0]
  spawn roll -0.029392 rad; rate -0.22538 rad/s
  gyro bias XYZ rad/s [0.000406189, -0.000215817, 4.80454e-05]
  gravity bias YZ [-0.00327204, -0.00162364]
Six significant digits shown; full precision: sampled_dr_values.json.
```

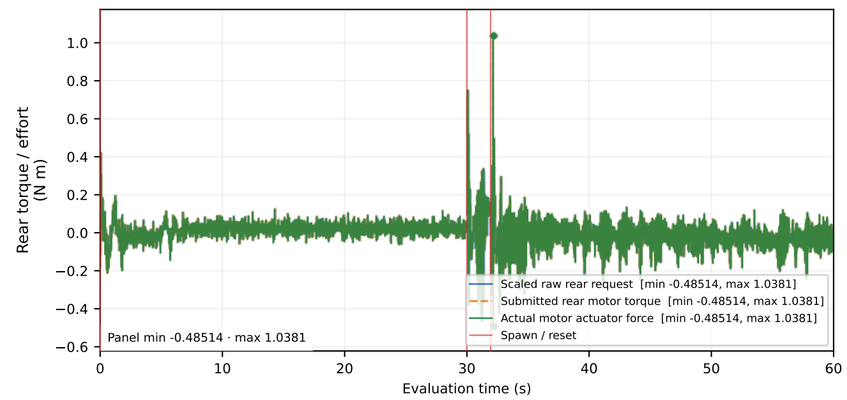
# model_2500 · Demand versus actuation · env 6 · full 60 s

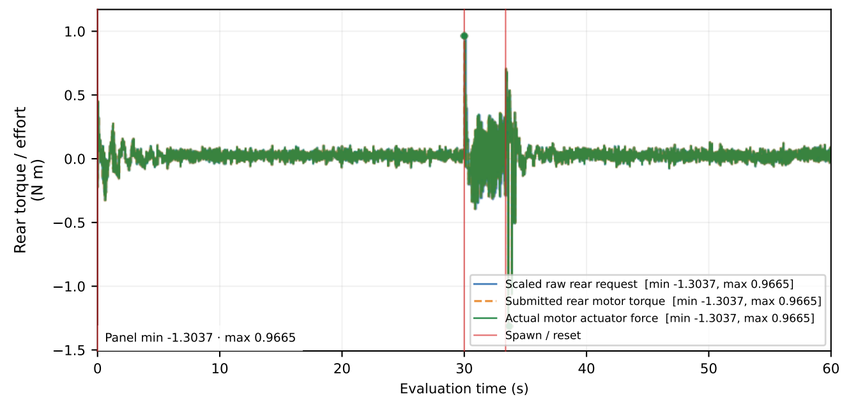


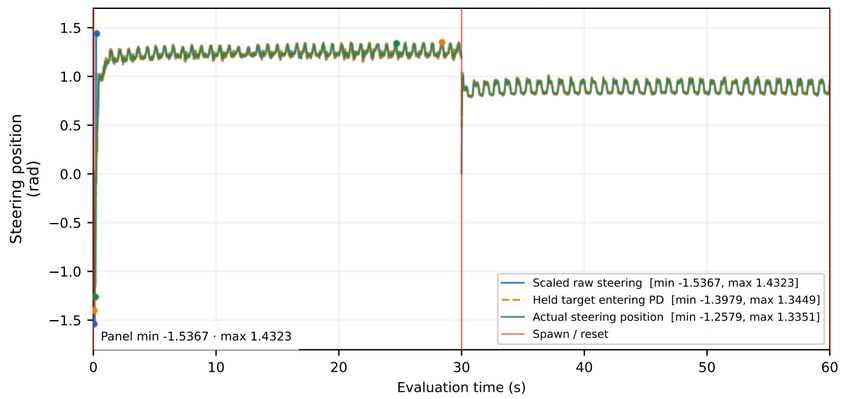

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

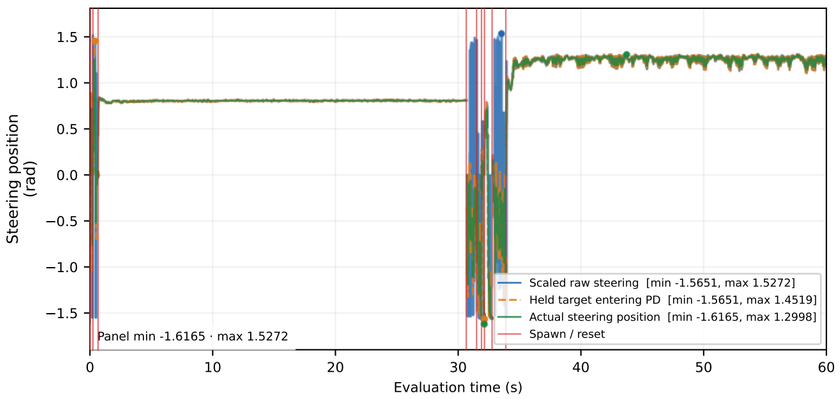
```
SAMPLED DR: env 6; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.878882, 0.980103, 0.820446, 0.995684]
  CoM XYZ mm [4.88032, 1.28894, -0.144839]
  mu_s [0.713129, 0.659445, 0.560535, 0.532645]
  mu_d [0.713129, 0.547619, 0.560535, 0.532645]
  Kp [steer,rear] [10.8556, 0]; Kd [0.211197, 0]
  spawn roll -0.0759049 rad; rate 0.0792792 rad/s
  gyro bias XYZ rad/s [-0.000271052, 0.000281376, 0.00032944]
  gravity bias YZ [0.000762568, -0.00113228]
E2 [30.00, 60.00) s:
  mass x [1.16766, 1.02214, 1.16892, 1.14839]
  CoM XYZ mm [4.2415, -2.19653, 4.4646]
  mu_s [0.547743, 0.628821, 0.579974, 0.592046]
  mu_d [0.547743, 0.628821, 0.579974, 0.534947]
  Kp [steer,rear] [9.40059, 0]; Kd [0.215543, 0]
  spawn roll -0.0541233 rad; rate 0.0767112 rad/s
  gyro bias XYZ rad/s [0.00019548, -1.27438e-05, 0.000412792]
  gravity bias YZ [-0.00398889, -0.000862655]
Six significant digits shown; full precision: sampled_dr_values.json.
```

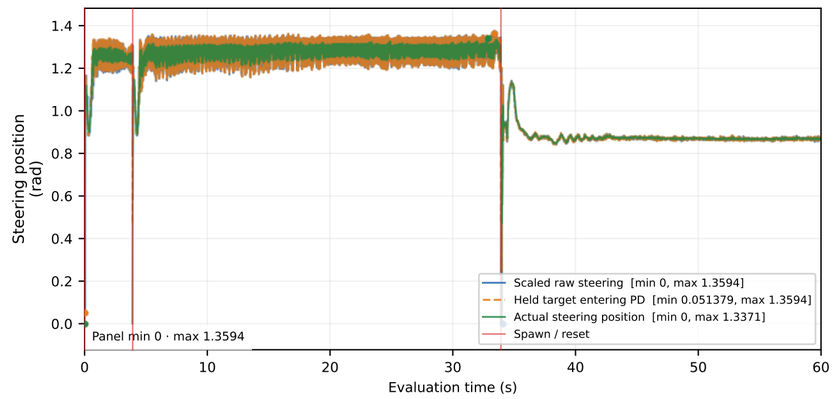
```
SAMPLED DR: env 6; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.95508, 1.03831, 1.00552, 1.0778]
  CoM XYZ mm [1.52497, 3.63117, 0.89843]
  mu_s [0.538434, 0.597748, 0.559972, 0.704749]
  mu_d [0.538434, 0.597748, 0.502209, 0.541926]
  Kp [steer,rear] [9.42108, 0]; Kd [0.183421, 0]
  spawn roll 0.0479951 rad; rate -0.0975409 rad/s
  gyro bias XYZ rad/s [-7.63979e-06, -0.000148874, 0.000268199]
  gravity bias YZ [0.0011309, 0.00172192]
E2 [30.00, 31.94) s:
  mass x [1.0515, 1.19247, 1.19836, 1.13004]
  CoM XYZ mm [4.57824, -4.74443, -1.52432]
  mu_s [0.559972, 0.631146, 0.757579, 0.732187]
  mu_d [0.502209, 0.631146, 0.70921, 0.631664]
  Kp [steer,rear] [9.23497, 0]; Kd [0.200202, 0]
  spawn roll 0.0695217 rad; rate -0.141358 rad/s
  gyro bias XYZ rad/s [-0.00028202, 0.000293422, 0.000191959]
  gravity bias YZ [1.23925e-05, 0.00172226]
E3 [31.94, 60.00) s:
  mass x [1.07042, 0.980495, 0.984247, 1.02319]
  CoM XYZ mm [-0.26196, -4.59484, 2.13633]
  mu_s [0.616244, 0.727426, 0.640016, 0.71155]
  mu_d [0.586498, 0.606358, 0.513141, 0.71155]
  Kp [steer,rear] [9.82772, 0]; Kd [0.219327, 0]
  spawn roll -0.00767592 rad; rate -0.166608 rad/s
  gyro bias XYZ rad/s [8.75006e-06, -0.000253532, 0.000100475]
  gravity bias YZ [-0.0003994, 2.07428e-05]
Six significant digits shown; full precision: sampled_dr_values.json.
```

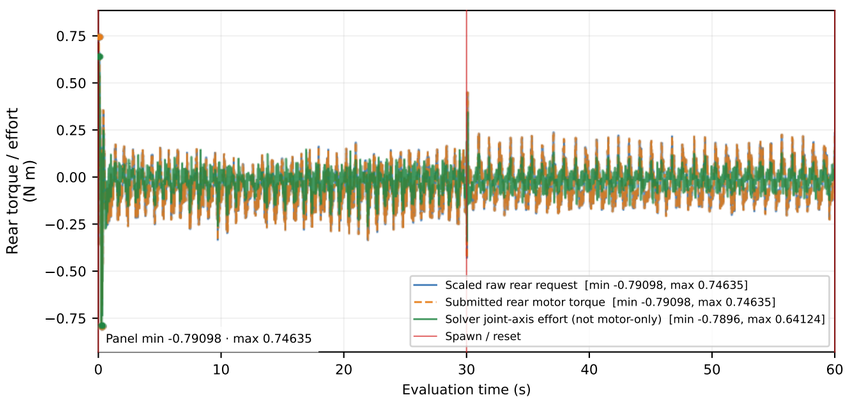
```
SAMPLED DR: env 6; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.95508, 1.03831, 1.00552, 1.0778]
  CoM XYZ mm [1.52497, 3.63117, 0.89843]
  mu_s [0.538434, 0.597748, 0.559972, 0.704749]
  mu_d [0.538434, 0.597748, 0.502209, 0.541926]
  Kp [steer,rear] [9.42108, 0]; Kd [0.183421, 0]
  spawn roll 0.0479951 rad; rate -0.0975409 rad/s
  gyro bias XYZ rad/s [-7.63979e-06, -0.000148874, 0.000268199]
  gravity bias YZ [0.0011309, 0.00172192]
E2 [30.00, 33.38) s:
  mass x [1.0515, 1.19247, 1.19836, 1.13004]
  CoM XYZ mm [4.57824, -4.74443, -1.52432]
  mu_s [0.559972, 0.631146, 0.757579, 0.732187]
  mu_d [0.502209, 0.631146, 0.70921, 0.631664]
  Kp [steer,rear] [9.23497, 0]; Kd [0.200202, 0]
  spawn roll 0.0695217 rad; rate -0.141358 rad/s
  gyro bias XYZ rad/s [-0.00028202, 0.000293422, 0.000191959]
  gravity bias YZ [1.23925e-05, 0.00172226]
E3 [33.38, 60.00) s:
  mass x [1.12701, 0.816186, 1.01265, 1.19488]
  CoM XYZ mm [-0.710368, 3.23515, 2.72403]
  mu_s [0.669571, 0.700521, 0.748289, 0.538976]
  mu_d [0.669571, 0.641329, 0.689499, 0.538976]
  Kp [steer,rear] [10.8693, 0]; Kd [0.190503, 0]
  spawn roll -0.0604354 rad; rate -0.128044 rad/s
  gyro bias XYZ rad/s [5.34909e-05, 0.000369712, 0.00025429]
  gravity bias YZ [0.00398337, 0.00196253]
Six significant digits shown; full precision: sampled_dr_values.json.
```

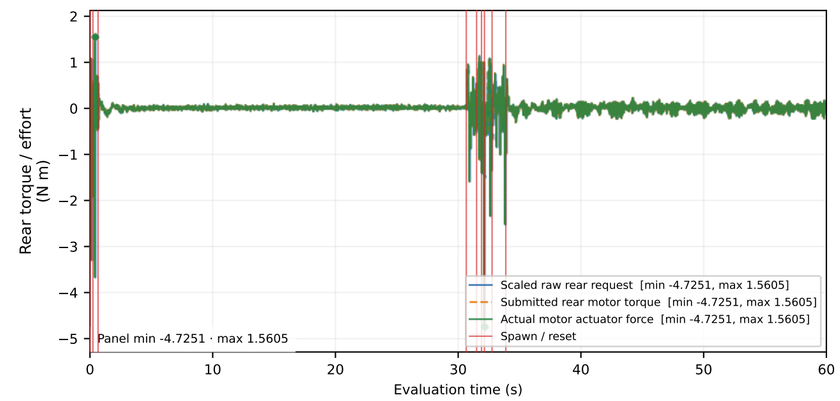
# model_2500 · Demand versus actuation · env 7 · full 60 s

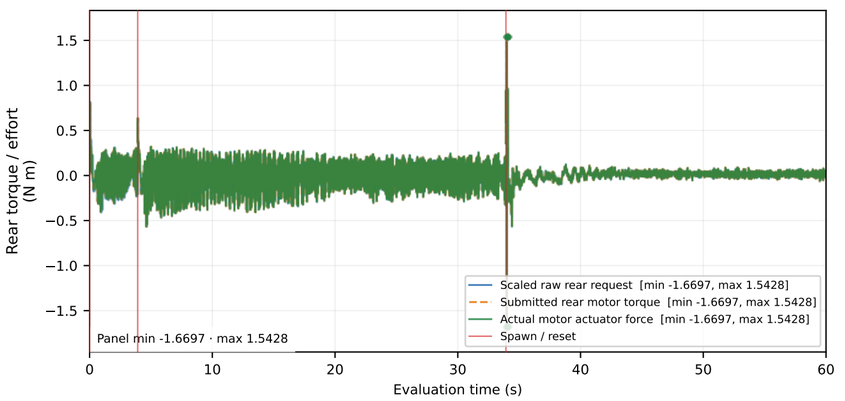



50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Green torque: Isaac solver joint-axis effort ≠ isolated motor torque; MuJoCo actuator force ≠ net joint torque.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 7; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.17324, 0.952299, 0.82034, 0.906501]
  CoM XYZ mm [2.37444, -2.24125, 4.08924]
  mu_s [0.794338, 0.547743, 0.547743, 0.547743]
  mu_d [0.526209, 0.547743, 0.547743, 0.547743]
  Kp [steer,rear] [9.28848, 0]; Kd [0.203181, 0]
  spawn roll -0.093371 rad; rate 0.204639 rad/s
  gyro bias XYZ rad/s [0.000258929, 0.000124329, 6.37829e-05]
  gravity bias YZ [0.000248948, 0.00114776]
E2 [30.00, 60.00) s:
  mass x [1.12148, 1.1744, 1.19594, 1.05127]
  CoM XYZ mm [4.01582, 3.58083, -0.0416934]
  mu_s [0.681711, 0.760821, 0.647391, 0.681711]
  mu_d [0.611756, 0.670315, 0.647391, 0.611756]
  Kp [steer,rear] [10.4243, 0]; Kd [0.195552, 0]
  spawn roll 0.0374949 rad; rate 0.110923 rad/s
  gyro bias XYZ rad/s [-0.000182822, -9.52629e-05, 0.000164718]
  gravity bias YZ [0.00459708, -0.000579513]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 7; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.24) s:
  mass x [0.945142, 1.03729, 0.95678, 1.04948]
  CoM XYZ mm [1.55815, -4.86414, 2.76336]
  mu_s [0.748289, 0.669571, 0.732187, 0.597748]
  mu_d [0.689499, 0.669571, 0.631664, 0.597748]
  Kp [steer,rear] [9.22333, 0]; Kd [0.205043, 0]
  spawn roll 0.0579807 rad; rate -0.230146 rad/s
  gyro bias XYZ rad/s [0.00021898, 0.000178088, 0.000100494]
  gravity bias YZ [-0.000466871, -0.000566274]
E2 [0.24, 0.66) s:
  mass x [0.815334, 0.846732, 1.04527, 0.935875]
  CoM XYZ mm [-0.394532, -2.53399, 4.82514]
  mu_s [0.727426, 0.704749, 0.541939, 0.541939]
  mu_d [0.606358, 0.541926, 0.534359, 0.534359]
  Kp [steer,rear] [9.38754, 0]; Kd [0.200427, 0]
  spawn roll 0.0472496 rad; rate 0.205943 rad/s
  gyro bias XYZ rad/s [-4.53926e-05, -0.000374891, 0.000302147]
  gravity bias YZ [-0.000488244, -0.000904653]
E3 [0.66, 30.66) s:
  mass x [1.05443, 1.07439, 1.18194, 1.1361]
  CoM XYZ mm [1.26006, 4.69445, 2.3054]
  mu_s [0.637675, 0.568073, 0.748289, 0.732187]
  mu_d [0.637675, 0.568073, 0.689499, 0.631664]
  Kp [steer,rear] [10.13, 0]; Kd [0.180181, 0]
  spawn roll 0.0459918 rad; rate -0.0558392 rad/s
  gyro bias XYZ rad/s [0.000193141, -0.000186143, 3.18235e-05]
  gravity bias YZ [-0.00166348, -0.000338645]
E4 [30.66, 31.50) s:
  mass x [1.01205, 1.11564, 1.17844, 1.01406]
  CoM XYZ mm [2.38007, 4.25924, 1.85643]
  mu_s [0.748289, 0.757579, 0.728342, 0.723429]
  mu_d [0.689499, 0.70921, 0.728342, 0.723429]
  Kp [steer,rear] [9.87057, 0]; Kd [0.191116, 0]
  spawn roll -0.0839131 rad; rate -0.0139192 rad/s
  gyro bias XYZ rad/s [9.68566e-05, 0.00027244, 5.12429e-05]
  gravity bias YZ [0.00121731, 0.000457613]
E5 [31.50, 31.90) s:
  mass x [0.844378, 1.19574, 0.87344, 1.11659]
  CoM XYZ mm [1.87176, 1.99566, -3.27129]
  mu_s [0.728342, 0.640016, 0.69316, 0.69316]
  mu_d [0.728342, 0.513141, 0.69316, 0.69316]
  Kp [steer,rear] [9.49388, 0]; Kd [0.188026, 0]
  spawn roll -0.0894517 rad; rate 0.133838 rad/s
  gyro bias XYZ rad/s [0.000129528, -0.000337925, 1.56767e-06]
  gravity bias YZ [-0.000239414, 0.00301222]
E6 [31.90, 32.14) s:
  mass x [0.967987, 1.12543, 1.13584, 0.841162]
  CoM XYZ mm [-0.510682, 0.130745, -2.72529]
  mu_s [0.546287, 0.559972, 0.71155, 0.727426]
  mu_d [0.546287, 0.502209, 0.71155, 0.606358]
  Kp [steer,rear] [9.57272, 0]; Kd [0.217513, 0]
  spawn roll -0.081501 rad; rate 0.0404057 rad/s
  gyro bias XYZ rad/s [4.89832e-05, -1.75448e-05, 3.42075e-05]
  gravity bias YZ [-0.000777028, -0.00227761]
E7 [32.14, 32.76) s:
  mass x [1.01407, 1.17122, 1.02528, 1.13515]
  CoM XYZ mm [-2.13715, 4.81493, -4.6917]
  mu_s [0.541939, 0.568073, 0.757579, 0.538434]
  mu_d [0.534359, 0.568073, 0.70921, 0.538434]
  Kp [steer,rear] [10.7901, 0]; Kd [0.202416, 0]
  spawn roll -0.0138355 rad; rate 0.236871 rad/s
  gyro bias XYZ rad/s [-0.000234231, -0.000145991, -0.000395346]
  gravity bias YZ [-0.00228167, -0.00111487]
E8 [32.76, 33.88) s:
  mass x [0.983052, 1.13748, 1.1842, 0.97888]
  CoM XYZ mm [-2.31369, 2.93858, 0.875108]
  mu_s [0.640016, 0.71008, 0.559972, 0.528253]
  mu_d [0.513141, 0.59371, 0.502209, 0.528253]
  Kp [steer,rear] [9.09864, 0]; Kd [0.206216, 0]
  spawn roll -0.0149822 rad; rate 0.0646525 rad/s
  gyro bias XYZ rad/s [0.000110128, -0.000243677, -0.000276047]
  gravity bias YZ [0.00222282, 0.000648969]
E9 [33.88, 60.00) s:
  mass x [1.15101, 1.0811, 1.0659, 1.14892]
  CoM XYZ mm [-4.29222, -1.57093, 1.5318]
  mu_s [0.631146, 0.704749, 0.528253, 0.633024]
  mu_d [0.631146, 0.541926, 0.528253, 0.568172]
  Kp [steer,rear] [10.9784, 0]; Kd [0.188898, 0]
  spawn roll 0.0314797 rad; rate -0.0384315 rad/s
  gyro bias XYZ rad/s [-0.000283127, -6.42477e-06, -0.000253066]
  gravity bias YZ [0.00249299, 0.00163119]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 7; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 3.92) s:
  mass x [0.945142, 1.03729, 0.95678, 1.04948]
  CoM XYZ mm [1.55815, -4.86414, 2.76336]
  mu_s [0.748289, 0.669571, 0.732187, 0.597748]
  mu_d [0.689499, 0.669571, 0.631664, 0.597748]
  Kp [steer,rear] [9.22333, 0]; Kd [0.205043, 0]
  spawn roll 0.0579807 rad; rate -0.230146 rad/s
  gyro bias XYZ rad/s [0.00021898, 0.000178088, 0.000100494]
  gravity bias YZ [-0.000466871, -0.000566274]
E2 [3.92, 33.92) s:
  mass x [1.19354, 0.806697, 1.18383, 1.05628]
  CoM XYZ mm [4.05146, -3.03807, 0.596929]
  mu_s [0.640016, 0.749678, 0.736077, 0.71155]
  mu_d [0.513141, 0.741429, 0.699455, 0.71155]
  Kp [steer,rear] [10.8219, 0]; Kd [0.206327, 0]
  spawn roll 0.0704726 rad; rate 0.0819431 rad/s
  gyro bias XYZ rad/s [-0.000146428, -2.8178e-05, 0.000313634]
  gravity bias YZ [0.00083648, -0.000763304]
E3 [33.92, 60.00) s:
  mass x [0.933955, 1.15257, 0.863114, 0.810173]
  CoM XYZ mm [-0.998784, 3.44166, 4.94422]
  mu_s [0.736077, 0.616244, 0.727426, 0.640016]
  mu_d [0.699455, 0.586498, 0.606358, 0.513141]
  Kp [steer,rear] [10.2999, 0]; Kd [0.21851, 0]
  spawn roll 0.087129 rad; rate 0.0232131 rad/s
  gyro bias XYZ rad/s [0.000176006, -0.00013264, 7.32382e-05]
  gravity bias YZ [-0.000695073, 0.00255973]
Six significant digits shown; full precision: sampled_dr_values.json.
```